

Process Characterization Report

A-Mab Drug Substance — Cation Exchange Chromatography (Step 7) — PCR-007

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—

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Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion-exchange chromatography · ANOVA, analysis of variance · BLA, Biologics License Application · CCD, central-composite design · CEX, cation-exchange chromatography · CPP, critical process parameter · CQA, critical quality attribute · CV, column volume · DoE, design of experiments · DS, drug substance · ELISA, enzyme-linked immunosorbent assay · GPP, general process parameter · HCP, host-cell protein · HMW, high-molecular-weight (aggregate) · ICH, International Council for Harmonisation · IPC, in-process control · KPP, key process parameter · MSAT, Manufacturing Science and Technology · NOR, normal operating range · PAR, proven acceptable range · PPQ, process performance qualification · QbD, Quality by Design · RSM, response-surface methodology · SDM, scale-down model · SEC, size-exclusion chromatography · WC-CPP, well-controlled CPP.

Executive summary

This report documents the Stage 1 (Process Design) characterization of the A-Mab cation-exchange (CEX) polishing chromatography step (Step 7), the bind-and-elute polishing operation that follows low-pH viral inactivation and is the **principal aggregate-reduction step** of the purification train — the only step that actively reduces high-molecular-weight aggregate — as well as a major clearance step for host-cell protein (HCP) and a modular clearance step for residual DNA and leached Protein A. The study was executed under the Process Characterization Plan **PCP-007** on a qualified scale-down chromatography column, following the lifecycle approach to process validation (U.S. Food and Drug Administration 2011; Parenteral Drug Association 2013; International Society for Pharmaceutical Engineering 2023) and the Quality-by-Design principles of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012), with the process model, quality-attribute framework and risk methodology of the A-Mab case study (CMC Biotech Working Group 2009).

Five process parameters were evaluated. A 19-run two-level full factorial screening design identified the significant factors and two-factor interactions for the eluate-pool aggregate, the eluate-pool HCP and the step yield; a 28-run face-centred central-composite response-surface design then quantified main effects, interactions and curvature and defined the multivariate operating region in the well-controlled CPPs. The aggregate and pool-HCP response-surface models are adequate and predictive ($R^2 = 0.97\text{--}0.98$, adjusted $R^2 = 0.93$, predicted $R^2 = 0.82$); the step yield is governed by protein load and is otherwise robust.

Key outcomes

- All four multivariate parameters significantly affect a product-quality CQA and are classified **well-controlled CPP (WC-CPP)**: protein load, elution pH and the elution stop-collect drive the eluate-pool aggregate, and the load/wash conductivity and protein load drive the eluate-pool HCP through a significant load $\times$ conductivity interaction. The elution flow rate has no product-quality or performance impact over its range and is **GPP**; there are no key process parameters (KPP) at this step.
- Aggregate rises with higher protein load, higher elution pH and a later stop-collect and is insensitive to the wash conductivity; the step reduces the pool aggregate by approximately **2.6-fold** to **0.73% HMW** at the nominal condition, well below the 5% drug-substance limit.
- HCP clearance improves with higher load/wash conductivity and falls at high load; the step reduces the pool HCP by approximately **78-fold** to about **117 ng/mg** at the nominal condition, which the anion-exchange step further polishes to the 100 ng/mg drug-substance limit.
- The nominal step recovers **95.2%** of the loaded product; yield declines modestly at high load and is otherwise robust.
- A multivariate operating region in the four well-controlled CPPs was established over which the eluate-pool aggregate and HCP are controlled so that the integrated control strategy delivers in-specification drug substance, with a commercial-scale aggregate capability of **Cpk = 16.11** — the step being the principal contributor to the drug-substance aggregate outcome.

The cation-exchange step is demonstrated to be well understood and robust. The operating region, parameter classifications and proven acceptable ranges support the Stage 2 operating ranges and the drug-substance control strategy, and this report rolls up into the Process Characterization Master Report (**PCMR-001**).

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody purified by a platform Protein A → low-pH viral-inactivation → cation-exchange → anion-exchange → virus-filtration → UF/DF train (CMC Biotech Working Group 2009). Cation-exchange chromatography is the first polishing step: the neutralized viral-inactivation pool is loaded onto a strong cation-exchange resin at controlled conductivity, weakly bound impurities are removed in the wash, and the product is recovered by elution, with the eluate pool collected between defined start and stop criteria. The step forms no new product-quality CQA; its role in the control strategy is to reduce the high-molecular-weight aggregate carried into the pool — including the aggregate that can arise during the upstream low-pH hold — and to provide a major further reduction of HCP and modular clearance of residual DNA and leached Protein A. This characterization quantifies how the polishing parameters govern the eluate-pool aggregate, the eluate-pool HCP and the step yield.

1.2 Objectives

The objectives of the study were to (i) establish the quantitative relationships between the cation-exchange process parameters and the eluate-pool aggregate, the eluate-pool HCP and the step yield; (ii) define the multivariate operating region over which the aggregate and HCP presented to the anion-exchange step are controlled; (iii) establish proven acceptable ranges (PARs) and normal operating ranges (NORs) for each parameter; (iv) classify each parameter for its risk of impact to CQAs and to process performance; and (v) provide the process understanding and ranges that justify the Stage 2 operating conditions and the control strategy.

1.3 Regulatory and scientific basis

The work follows the lifecycle approach to process validation of the FDA 2011 guidance (U.S. Food and Drug Administration 2011), PDA Technical Report 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023), and applies the enhanced (QbD) development approach of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012). Criticality is treated as a continuum (ICH Q9). The report structure corresponds to the Stage 1 Process Design Report of PDA TR 60 §3.11 and provides development-report content suitable to support a Biologics License Application.

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Cation-exchange polishing is a platform operation whose behaviour is well established from related humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and from A-Mab clinical and engineering experience (CMC Biotech Working Group 2009). Prior knowledge established that aggregate, being more highly charged and more strongly retained than monomer, is resolved from monomer in the elution so that the eluate-pool aggregate rises with higher protein load, with a later stop-collect (which extends the pool further down the descending edge into the late-eluting aggregate) and at higher elution pH; that HCP clearance is governed by the load/wash conductivity — a higher conductivity strips more weakly bound HCP before the elution — and falls at high load, the two acting together through a load $\times$ conductivity interaction; and that recovery declines modestly at high load. This knowledge framed the attributes in scope and the prioritization of parameters for characterization.

2.2 Quality attributes in scope

Cation exchange introduces no CQA. It is the principal reduction step for the high-molecular-weight aggregate and a major clearance step for the impurity CQAs formed upstream and specified at the drug substance: host-cell protein, residual DNA and leached Protein A (Table 2.1). Criticality is scored on the A-Mab continuum using Tool #1 (Risk Score = Impact $\times$ Uncertainty); attributes rated High (H) or Very High (VH) are treated as *Critical* (CMC Biotech Working Group 2009).

Table 2.1: CQAs controlled by the cation-exchange step (formed upstream; polished/cleared here), with acceptance criterion and criticality (Tool #1 = Impact $\times$ Uncertainty).

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16

The eluate-pool aggregate and eluate-pool HCP are not themselves released specifications; they are in-process attributes whose control at this step determines the size-variant and impurity load presented to the anion-exchange step and therefore the margin to the 5% HMW aggregate and 100 ng/mg HCP drug-substance limits. Because aggregate can be elevated in the incoming pool, the study was designed to demonstrate the aggregate-clearance capacity of the step against a

representative, deliberately challenging aggregate feed; pool aggregate is characterized here as the primary response of the study.

2.3 Risk-based prioritization and parameter selection

Consistent with the A-Mab lifecycle of iterative risk assessments, the characterization scope was set by the Pre-Characterization Process Risk Assessment (**RA-001**), which used prior knowledge and a risk-ranking-and-filtering screen to prioritize parameters and to assign the study type. Protein load, load/wash conductivity, elution pH and the elution stop-collect, which carry a credible main-effect and interaction risk to the eluate-pool aggregate or HCP, were assigned to the multivariate DoE; the elution flow rate, expected to affect only process performance over a wide range, was assigned to univariate assessment (CMC Biotech Working Group 2009; Parenteral Drug Association 2013). The parameters and ranges carried into this study are given in Table 4.1 (§5.1).

3 Materials and methods

3.1 Scale-down model and its qualification

All characterization runs were performed on a qualified scale-down cation-exchange column operated at the commercial-process equivalents for resin, bed height, residence time, buffer system, load ratio and elution strategy (**SOP-2010**, **SOP-2008**). The SDM was qualified against at-scale reference data (**SOP-1001**) by comparing the operational parameters and, critically, the input and output attributes — step yield, eluate-pool aggregate and HCP, residual DNA and leached Protein A — confirming that the model is representative and suitable to support operating-range and PAR claims (Parenteral Drug Association 2013). The between-run reproducibility of the centre-point condition (§6.1) provides a direct estimate of model precision and is used as the pure-error term in the lack-of-fit assessment of every response-surface model.

3.2 Operation

The neutralized viral-inactivation pool is conditioned to the target load conductivity and loaded onto the equilibrated cation-exchange column to the target load ratio; the column is washed at the load/wash conductivity to remove weakly bound host-cell protein and other impurities; and the product is recovered by elution, with the eluate pool collected from a defined start until the descending-edge stop-collect criterion. Buffers, resin lot and column packing are controlled to platform specifications (**SOP-2008**) and are treated as identity/quality-controlled inputs rather than characterized variables, consistent with the A-Mab platform approach (CMC Biotech Working Group 2009).

3.3 Analytical methods

The step responses were measured by validated methods (Table 3.1). Aggregate was determined by SEC-HPLC (**AMV-3011**), the primary assay for the aggregate-polish claim; host-cell protein by ELISA (**AMV-3012**); residual host-cell DNA by qPCR (**AMV-3014**); and leached Protein A by ELISA (**AMV-3016**). Step yield was determined from the load and eluate product mass by protein A . Method performance characteristics are summarized in Appendix D.

Table 3.1: Validated analytical methods used in the characterization.

Reference	Title	Type
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation

Reference	Title	Type
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation

3.4 Sampling plan

Each DoE run was sampled at the eluate pool for aggregate, HCP, residual DNA and leached Protein A, and the load and pool product mass were recorded for the step-yield determination; in-process readings (pH, conductivity, UV) confirmed the chromatographic profile and the pool collection window. Centre-point runs were replicated within each design (three in screening, four in the response-surface design) to quantify reproducibility and pure error.

3.5 Statistical methods

Designs were constructed and analyzed in coded factors, with each parameter’s characterization range mapped so that the set-point is 0 and the range edges are ± 1 (**SOP-1002**). Screening data were analyzed by ordinary-least-squares regression of each response on the main effects and all two-factor interactions; factor *effects* are reported as twice the coded coefficient. Response-surface data were analyzed with the full quadratic model (main effects, two-factor interactions and pure-quadratic terms). Statistical significance was assessed at $\alpha = 0.05$. Model adequacy was judged by the coefficient of determination (R^2), the adjusted R^2 , the predicted R^2 (computed from the PRESS statistic by leave-one-out), the overall model F-test, and an analysis of variance partitioning the residual into lack of fit and pure error, the latter estimated from the replicated centre points; a non-significant lack-of-fit F-test ($p > 0.05$) indicates an adequate model. A response governed by a single dominant factor is reported by that dependence and treated as otherwise robust rather than as a full predictive surface. The operating region was defined as the multivariate region of the well-controlled CPPs over which the eluate-pool aggregate and HCP are controlled. Commercial-scale process capability for the drug-substance CQAs governed by the step was estimated by Monte-Carlo simulation of 2,000 batches with each parameter varied within its NOR, reporting the one-sided process-capability index (Cpk) for the impurity and size-variant CQAs.

4 Study design

4.1 Factors, ranges and the knowledge space

Five parameters were studied (Table 4.1). The characterization range of each factor (“PAR” column) defines the edges of the explored knowledge space and the basis of the proven acceptable range; the NOR is the tighter range at which the parameter is controlled routinely. The DoE factor coding for the four multivariate factors is given in Table 4.2.

Table 4.1: Cation-exchange process parameters, set-points, ranges and post-characterization classification.

Parameter	Unit	Set-point	NOR	PAR	Class	Study
Protein load	g/L resin	25	18–32	10–40	WC-CPP	multivariate
Load/Wash conductivity	mS/cm	5	4–6	3–7	WC-CPP	multivariate
Elution buffer pH	pH	6	5.85–6.15	5.8–6.2	WC-CPP	multivariate
Elution stop collect	OD desc.	1	0.7–1.3	0.5–1.5	WC-CPP	multivariate
Elution flow rate	cm/hr	200	150–250	100–300	GPP	univariate

Table 4.2: Design-of-experiments factor legend (coded A–D).

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Load/Wash conductivity	mS/cm	screening + RSM
C	Elution buffer pH	pH	screening + RSM
D	Elution stop collect	OD desc.	screening + RSM

4.2 Screening design

Factor screening used a two-level full factorial in the four multivariate factors (A–D) with three centre-point replicates, 19 runs in total. The full factorial estimates all four main effects and all six two-factor interactions without confounding. The three responses recorded for each run were the eluate-pool aggregate, the eluate-pool HCP and the step yield. The complete coded design matrix and run responses are given in Appendix A.

4.3 Response-surface design

The four multivariate factors were carried into a face-centred central-composite design (cube points, axial points at the face centres, and four centre-point replicates), 28 runs in total, enabling estimation of curvature in addition to main effects and interactions. The complete coded design matrix and responses are given in Appendix B.

4.4 Univariate assessment

The elution flow rate was evaluated one factor at a time across its characterization range for its effect on step yield, the eluate-pool aggregate and HCP, and the chromatographic profile; it was not carried into the multivariate design because prior knowledge indicated no credible impact to a product-quality CQA.

5 Results

5.1 Centre-point performance and reproducibility

At the nominal (centre-point) condition the step recovers 95.2% of the loaded product and delivers an eluate-pool aggregate of approximately 0.77% HMW and an eluate-pool HCP of approximately 134 ng/mg. The between-replicate reproducibility of the centre point (Table 5.1) is the pure-error estimate used in the lack-of-fit assessment; the coefficients of variation reflect the combined analytical and process variability of the scale-down model, with the pool-HCP measurement the least precise of the three responses.

Table 5.1: Centre-point reproducibility (response-surface design; n = 4 replicates).

Response	n	Mean	SD	%CV
Aggregates (HMW, %)	4	0.771	0.043	5.58
Pool HCP (ng/mg)	4	134	18.7	13.9
Step yield	4	0.918	0.0289	3.15

5.2 Screening: factor effects

The screening models resolve the active factors and interactions for the eluate-pool aggregate and the eluate-pool HCP, and identify protein load as the single dominant factor for the step yield (Table 5.2). Consistent with a screening analysis, the magnitudes are refined by the response-surface models (§6.3).

Table 5.2: Screening-design model summary by response (main effects + two-factor interactions).

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Aggregates (HMW, %)	19	0.973	0.939	0.869	28.5	3.57e-05	0.096
Pool HCP (ng/mg)	19	0.936	0.857	0.467	11.8	0.000937	35.4
Step yield	19	0.924	0.829	0.498	9.74	0.00181	0.0115

The largest effects are tabulated for the eluate-pool aggregate (Table 5.3) and the eluate-pool HCP (Table 5.4). Full effect tables for all three responses are given in Appendix C.

Table 5.3: Largest standardized effects on eluate-pool aggregate (screening; effect = $2 \times$ coded coefficient; A = load, B = wash conductivity, C = elution pH, D = stop-collect).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.599	0.299	0.024	12.5	1.6e-06	***
C	0.34	0.17	0.024	7.09	0.000103	***
D	0.319	0.16	0.024	6.65	0.000161	***
AC	0.196	0.0979	0.024	4.08	0.00354	**
AD	0.154	0.0772	0.024	3.22	0.0123	*
CD	0.131	0.0657	0.024	2.74	0.0256	*
B	-0.0185	-0.00924	0.024	-0.385	0.71	
BC	0.0152	0.00758	0.024	0.316	0.76	

Table 5.4: Largest standardized effects on eluate-pool HCP (screening).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-127	-63.3	8.85	-7.15	9.7e-05	***
A	116	58.2	8.85	6.58	0.000173	***
AB	-56.7	-28.4	8.85	-3.21	0.0125	*
BD	39.6	19.8	8.85	2.24	0.0557	
D	-25.8	-12.9	8.85	-1.46	0.182	
AD	-24.6	-12.3	8.85	-1.39	0.202	
CD	24.4	12.2	8.85	1.38	0.205	
BC	18.4	9.18	8.85	1.04	0.33	

Protein load, elution pH and the elution stop-collect are the active factors for the eluate-pool aggregate — the pool aggregate rises with each — with significant load $\times$ elution-pH and load $\times$ stop-collect interactions; the wash conductivity has no significant effect on aggregate. The eluate-pool HCP, by contrast, is governed by the load/wash conductivity and the protein load, with a significant load $\times$ conductivity interaction: HCP clearance improves at higher conductivity and worsens at higher load, and the interaction means the conductivity matters most when the column is heavily loaded. The step yield is governed by protein load alone (recovery declines modestly at higher load) and shows no other significant dependence. The aggregate and HCP responses therefore respond to a partly shared set of factors, which is why a multivariate operating region — rather than independent one-dimensional ranges — is used to assure the size-variant and impurity load presented downstream.

5.3 Response-surface models

The full quadratic response-surface models for the eluate-pool aggregate and the eluate-pool HCP are adequate and predictive (Table 5.5): $R^2 = 0.97\text{--}0.98$, adjusted $R^2 = 0.93$, and predicted $R^2 = 0.82$, indicating good predictive capability rather than overfitting. For both modelled responses the lack-of-fit F-test was not significant ($p > 0.05$) relative to the centre-point pure error, confirming model adequacy. The step-yield surface has a lower predicted R^2 because, apart from the dominant

protein-load main effect, its variation is at the level of the centre-point reproducibility rather than systematic structure; the yield is therefore reported by its load dependence and treated as otherwise robust across the ranges studied.

Table 5.5: Response-surface model summary by response (full quadratic model).

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Aggregates (HMW, %)	28	0.975	0.949	0.823	36.9	3.85e-08	0.0749
Pool HCP (ng/mg)	28	0.966	0.929	0.827	26.3	3.07e-07	16.8
Step yield	28	0.825	0.637	0.196	4.38	0.00573	0.0188

The fitted coefficients for the eluate-pool aggregate and the eluate-pool HCP are given in Table 5.6 and Table 5.7, and the analysis of variance for the eluate-pool aggregate — including the lack-of-fit/pure-error partition — in Table 5.8.

Table 5.6: Response-surface coefficients for eluate-pool aggregate (coded factors A = load, B = wash conductivity, C = elution pH, D = stop-collect).

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.751	0.0259	29	3.36e-13	***
A	0.28	0.0176	15.9	6.77e-10	***
B	0.0276	0.0176	1.57	0.142	
C	0.172	0.0176	9.75	2.41e-07	***
D	0.157	0.0176	8.88	7.05e-07	***
AB	0.00964	0.0187	0.515	0.615	
AC	0.096	0.0187	5.13	0.000193	***
AD	0.0851	0.0187	4.54	0.000551	***
BC	-0.00211	0.0187	-0.113	0.912	
BD	0.0137	0.0187	0.733	0.477	
CD	0.0694	0.0187	3.71	0.00262	**
A ²	0.0798	0.0466	1.71	0.111	
B ²	-0.00875	0.0466	-0.188	0.854	
C ²	0.0264	0.0466	0.566	0.581	
D ²	0.0677	0.0466	1.45	0.17	

Table 5.7: Response-surface coefficients for eluate-pool HCP.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	141	5.81	24.3	3.21e-12	***
A	52.5	3.96	13.3	6.21e-09	***
B	-49.7	3.96	-12.6	1.21e-08	***
C	4.14	3.96	1.05	0.315	
D	-2.13	3.96	-0.538	0.6	
AB	-16.3	4.2	-3.87	0.00194	**

Term	Coef.	Std. err.	t	p-value	Sig.
AC	-4.27	4.2	-1.02	0.328	
AD	-4.33	4.2	-1.03	0.321	
BC	-14	4.2	-3.33	0.00542	**
BD	0.547	4.2	0.13	0.898	
CD	-1.6	4.2	-0.381	0.71	
A ²	-16.4	10.5	-1.56	0.142	
B ²	12.9	10.5	1.24	0.238	
C ²	2.31	10.5	0.221	0.829	
D ²	10.1	10.5	0.968	0.351	

Table 5.8: Analysis of variance for the eluate-pool aggregate response-surface model, with lack-of-fit and pure-error partition.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	2.89	14	0.207	36.9	3.85e-08
Residual	0.0729	13	0.0056	nan	nan
Lack of fit	0.0673	10	0.00673	3.63	0.158
Pure error	0.00556	3	0.00185	nan	nan

The pool-HCP and aggregate response surfaces over protein load and wash conductivity are shown in Figure 5.1; the response surfaces for all three responses over protein load and elution pH are shown in Figure 5.2, and the model diagnostics for the aggregate and pool-HCP models in Figure 5.3 and Figure 5.4. The residual-versus-predicted and normal-quantile plots show no systematic structure or departure from normality, and the actual-versus-predicted plots lie on the identity line, supporting the use of the models for the operating-region definition.

Cation exchange (CEX) — response surfaces (load × wash conductivity)

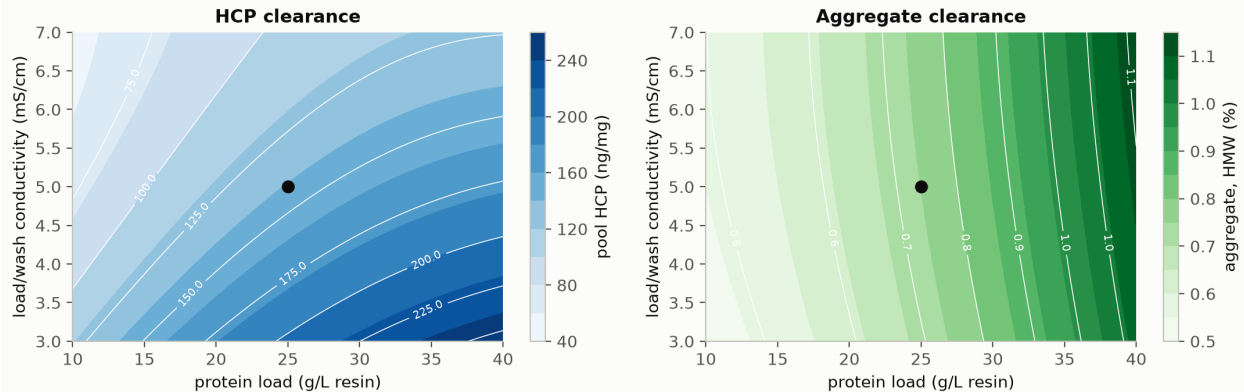


Figure 5.1: Cation-exchange response surfaces over protein load × wash conductivity: pool HCP falls with higher conductivity and rises with load through a significant load × conductivity interaction (left), while pool aggregate rises with load and is insensitive to conductivity (right). The set-point is marked.

Response-surface predictions (remaining factors held at set-point)

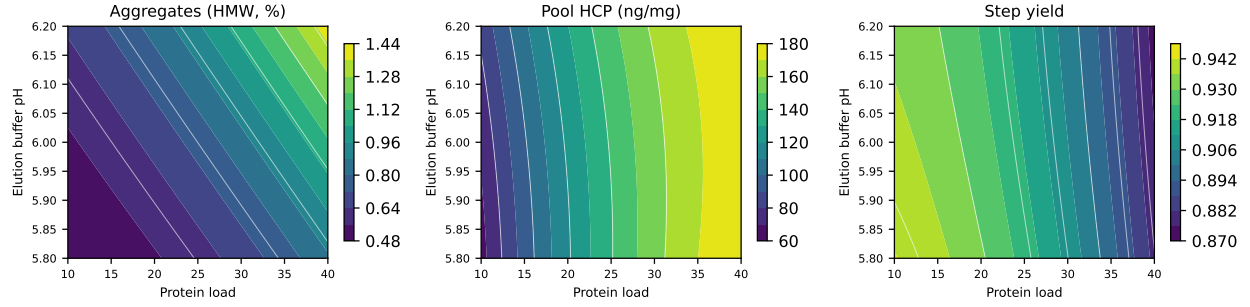


Figure 5.2: Response-surface predictions for the eluate-pool aggregate, eluate-pool HCP and step yield over protein load $\times$ elution pH (remaining factors at set-point). Aggregate rises with load and elution pH; HCP and yield respond to load.

Model diagnostics — Aggregates (HMW, %)

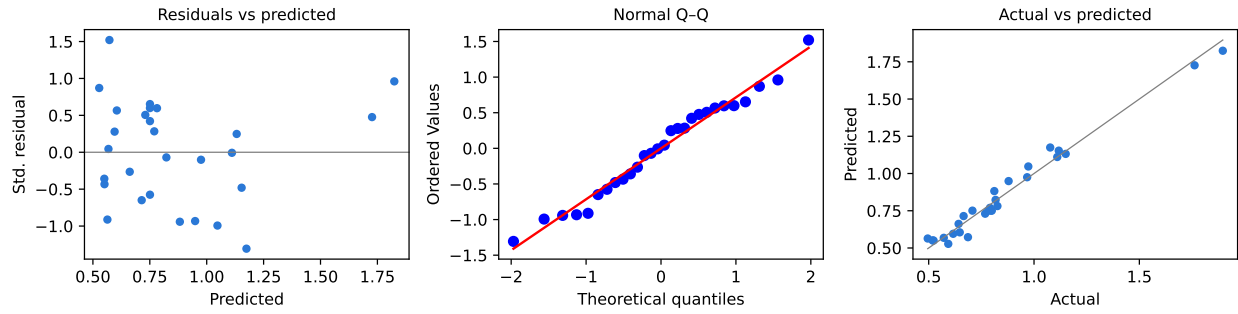


Figure 5.3: Model diagnostics for the eluate-pool aggregate response-surface model.

Model diagnostics — Pool HCP (ng/mg)

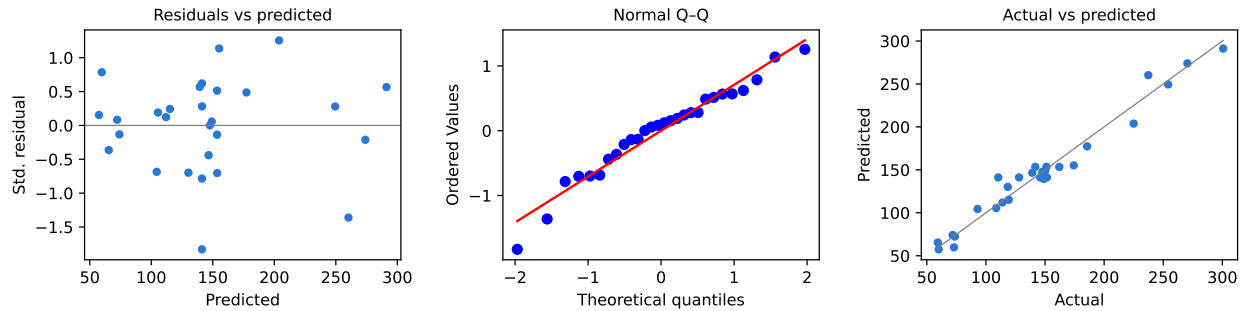


Figure 5.4: Model diagnostics for the eluate-pool HCP response-surface model.

5.4 Mechanistic interpretation

The response-surface models reproduce the established behaviour of cation-exchange polishing. High-molecular-weight aggregate carries a higher net positive charge and binds the resin more strongly than monomer, so it is resolved into the descending edge of the elution peak; the eluate-pool aggregate therefore rises when the column is more heavily loaded, when a higher elution pH weakens the resolution, and when the pool is collected further down the descending edge (a later stop-collect), with the load and pool-collection effects acting together through their interaction. The worst case for pool aggregate is a high load combined with a high elution pH and a late stop-collect. Host-cell protein, by contrast, is weakly retained and is removed principally in the wash: a higher load/wash conductivity displaces more HCP before the elution, improving clearance, while a heavier load reduces clearance, and the two act together through the significant load $\times$ conductivity interaction so that adequate wash conductivity matters most at high load. This is the relationship depicted in Figure 5.1 and is the reason a multivariate operating region — rather than independent one-dimensional ranges — is used to assure the aggregate and HCP presented downstream. The step yield declines modestly at high load, where breakthrough costs recovery, and is otherwise robust; the wash conductivity, which controls HCP, does not affect the aggregate, so the two clearance functions are governed largely by different parameters.

6 Design space

The operating region (design space) is the multivariate region of the four well-controlled CPPs over which the eluate-pool aggregate and HCP are held low enough that the drug substance meets its 5% HMW aggregate and 100 ng/mg HCP limits (the latter after the anion-exchange polish), while the step yield remains acceptable. Its two principal planes are protein load $\times$ wash conductivity, over which the pool HCP is controlled (Figure 5.1), and protein load $\times$ elution pH (with the stop-collect), over which the pool aggregate is controlled (Figure 5.2). The NOR of each parameter (Table 4.1) — protein load 18–32 g/L resin and load/wash conductivity 4–6 mS/cm, with elution pH and the stop-collect at their NORs — lies well inside the characterized region, which in turn lies within the broader knowledge space bounded by the proven acceptable ranges. Movement within the operating region is not a change, whereas movement outside it would require assessment under the quality system (CMC Biotech Working Group 2009; International Council for Harmonisation 2009). Because the step yield is robust and the two quality responses are governed by partly different parameters, the region is set jointly by the aggregate and HCP constraints.

7 Process capability and robustness

Cation exchange is the principal aggregate-reduction step of the process and a major clearance step for host-cell protein, residual DNA and leached Protein A. A Monte-Carlo simulation of 2,000 commercial-scale batches, with every parameter varied within its NOR, was used to estimate the commercial-scale distributions and process capability of the drug-substance CQAs to which this step contributes (Table 7.1). Capability is evaluated one-sided against the upper acceptance limit.

Table 7.1: Commercial-scale process capability for the CQAs governed by the cation-exchange step.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Aggregates (HMW)	H	upper	0–5	0.753 ± 0.088	16.11
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 ± 4.6	6.14
Residual DNA	VL	upper	0–0.001	0.0001 ± 0	10.81
Leached Protein A	L-M	upper	0–5	0.003 ± 0.0006	2785.30

The drug-substance aggregate meets its 5% HMW limit with a capability of **Cpk = 16.11**; as the only aggregate-reduction step, cation exchange is the primary contributor to this outcome. The drug-substance HCP meets its 100 ng/mg limit with a capability of Cpk = 6.14, to which this step contributes a major reduction that the anion-exchange step completes; and the residual DNA and leached Protein A are cleared to far below their limits (Cpk = 10.8 and 2785 respectively), with cation exchange providing modular clearance alongside the Protein A and anion-exchange steps. The capability margins confirm that the NOR-bounded step contributes robustly to in-specification drug substance at commercial scale.

8 Parameter classification

Each parameter was classified on the A-Mab continuum from the demonstrated effects and the operating-region risk (Table 8.1). The rationale is as follows.

- **Protein load — WC-CPP.** Significantly affects both the eluate-pool aggregate (its dominant factor) and the eluate-pool HCP, and participates in significant interactions with the elution pH, the stop-collect and the wash conductivity; it also governs the step yield. It is reliably held within the operating region by controlled load ratio, and is therefore a well-controlled CPP.
- **Load/wash conductivity — WC-CPP.** The dominant factor for HCP clearance (higher conductivity clears more HCP), with a significant load $\times$ conductivity interaction; it has no effect on aggregate. It is controlled by buffer preparation with in-process conductivity verification.
- **Elution pH, elution stop-collect — WC-CPP.** Each significantly affects the eluate-pool aggregate; both are reliably held within the operating region by controlled buffer preparation and a defined pool-collection criterion with in-process pH and UV monitoring.
- **Elution flow rate — GPP.** No meaningful impact on the CQAs or on performance was observed over the wide range studied.

Because every parameter that affects process performance (the step yield) also significantly affects a product-quality CQA, there are no key process parameters (KPP) at this step: the four multivariate parameters are well-controlled CPPs and the flow rate is a general process parameter.

Table 8.1: Post-characterization parameter classification.

Parameter	Class	Study
Protein load	WC-CPP	multivariate
Load/Wash conductivity	WC-CPP	multivariate
Elution buffer pH	WC-CPP	multivariate
Elution stop collect	WC-CPP	multivariate
Elution flow rate	GPP	univariate

9 Contribution to the control strategy

The cation-exchange step is the primary control point for high-molecular-weight aggregate and a major control point for the process-related impurity load. Its contribution to the overall control strategy comprises: control of the four WC-CPPs (protein load, load/wash conductivity, elution pH and the elution stop-collect) to their NORs within the established operating region; control of the elution flow rate as a general parameter; in-process monitoring of the load ratio, pH, conductivity and UV profile and of the pool-collection window; resin life-cycle control (**SOP-2008**); and release testing of the eluate-pool aggregate, HCP, residual DNA and leached Protein A. The aggregate that can form during the upstream low-pH hold is polished here, the host-cell protein reduced upstream by Protein A is further reduced here and completed by anion exchange, and the residual DNA and leached Protein A are cleared modularly across Protein A, cation exchange and anion exchange (reported in the paired unit-operation reports and consolidated in **PCMR-001**).

10 Discussion

The characterization quantifies the parameter–response relationships for the cation-exchange polishing step, defines a robust multivariate operating region in the four well-controlled CPPs, and classifies every parameter with a data-supported rationale. The aggregate and pool-HCP response-surface models are adequate and predictive, and their behaviour is consistent with established cation-exchange chromatography and with the A-Mab prior knowledge, increasing confidence that the SDM is representative of commercial scale. The step yield is governed by protein load and is otherwise robust. The principal limitation is inherent to scale-down characterization: the absolute aggregate and HCP clearance may shift modestly at scale and over the resin life-cycle, and the incoming aggregate load depends on the upstream low-pH hold, which is why the operating region is defined on the qualified SDM, challenged against a representative worst-case aggregate feed, confirmed at commercial scale during Stage 2, and carried forward with in-process monitoring and resin life-cycle control. The characterization did not identify any parameter requiring the most stringent (CPP, as opposed to WC-CPP) control within the studied ranges.

11 Conclusions

The A-Mab cation-exchange polishing step is well understood and robust. The relationships between the polishing parameters and the eluate-pool aggregate, the eluate-pool HCP and the step yield are quantified by DoE; a multivariate operating region in the four well-controlled CPPs is established; every parameter range is justified by data; and the drug-substance CQAs governed by the step meet acceptance at commercial scale, with the step being the principal contributor to the drug-substance aggregate capability of $Cpk = 16.11$. The operating region, parameter classifications and proven acceptable ranges provide the basis for the Stage 2 operating ranges and the drug-substance control strategy. This report rolls up into the Process Characterization Master Report (**PCMR-001**) and provides the parameter risk basis for the post-characterization process risk assessment.

Deviations from the plan

No deviations from **PCP-007** were recorded. All planned runs were executed and all planned responses were measured on the qualified scale-down model.

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Appendix A — Screening design matrix and responses

The coded two-level full-factorial design (factors A–D per Table 4.2; coded levels $-1/0/+1$) and the measured responses for each run.

Table 11.1: Appendix A1 — screening design (coded factor settings).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

Table 11.2: Appendix A2 — screening run responses.

Run	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
1	0.4997	136	0.9357
2	0.6038	114	0.9616
3	0.5478	135	0.9398
4	0.8177	138.2	0.9726
5	0.4518	66.23	0.934
6	0.5639	61.25	0.9509
7	0.5793	48.82	0.946

Run	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
8	0.7519	67.58	0.9344
9	0.7915	428.3	0.8827
10	1.097	246.9	0.8884
11	1.139	301.1	0.9227
12	1.788	239.6	0.9084
13	0.8065	105.3	0.895
14	1.036	112.6	0.8928
15	1.119	115.4	0.8844
16	1.829	149.3	0.8859
17	0.8333	152.9	0.929
18	0.829	182.9	0.9349
19	0.6919	128.7	0.9099

Appendix B — Response-surface design matrix and responses

The coded face-centred central-composite design (factors A–D; coded levels $-1/0/+1$) and the measured responses for each run.

Table 11.3: Appendix B1 — response-surface design (coded factor settings).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

Table 11.4: Appendix B2 — response-surface run responses.

Run	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
1	0.5185	108.7	0.9803
2	0.5929	114	0.9551
3	0.5716	151.2	0.9726
4	0.8176	141.8	0.9425
5	0.4954	59.31	0.9208
6	0.6162	71.9	0.9465
7	0.6864	60.06	0.9322
8	0.8115	72.97	0.9245
9	0.7681	237.4	0.8697
10	0.9729	254.1	0.8776
11	1.15	300.7	0.9004
12	1.762	270.4	0.8902
13	0.8262	174.3	0.9017
14	1.117	139.2	0.9032
15	1.077	118.4	0.8912
16	1.896	119.2	0.8437
17	0.5232	73.69	0.9353
18	1.11	185.6	0.8787
19	0.6657	225	0.9203
20	0.7908	92.9	0.9088
21	0.6474	148.9	0.9214
22	0.8793	147.7	0.9145
23	0.642	162.1	0.9347
24	0.9674	150.2	0.9217
25	0.7955	145.9	0.8811
26	0.7076	110.4	0.9517
27	0.7823	128	0.9207
28	0.7996	151.6	0.9173

Appendix C — Full effect and coefficient tables

11.1 Screening effects — all responses

Aggregates (HMW, %)

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.599	0.299	0.024	12.5	1.6e-06	***
C	0.34	0.17	0.024	7.09	0.000103	***
D	0.319	0.16	0.024	6.65	0.000161	***
AC	0.196	0.0979	0.024	4.08	0.00354	**
AD	0.154	0.0772	0.024	3.22	0.0123	*
CD	0.131	0.0657	0.024	2.74	0.0256	*
B	-0.0185	-0.00924	0.024	-0.385	0.71	
BC	0.0152	0.00758	0.024	0.316	0.76	
BD	-0.0131	-0.00657	0.024	-0.274	0.791	
AB	0.012	0.00602	0.024	0.251	0.808	

Pool HCP (ng/mg)

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-127	-63.3	8.85	-7.15	9.7e-05	***
A	116	58.2	8.85	6.58	0.000173	***
AB	-56.7	-28.4	8.85	-3.21	0.0125	*
BD	39.6	19.8	8.85	2.24	0.0557	
D	-25.8	-12.9	8.85	-1.46	0.182	
AD	-24.6	-12.3	8.85	-1.39	0.202	
CD	24.4	12.2	8.85	1.38	0.205	
BC	18.4	9.18	8.85	1.04	0.33	
AC	-12.5	-6.25	8.85	-0.707	0.5	
C	-9.45	-4.72	8.85	-0.534	0.608	

Step yield

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	-0.0518	-0.0259	0.00287	-9.04	1.79e-05	***
BC	-0.0121	-0.00607	0.00287	-2.12	0.0672	
B	-0.0111	-0.00553	0.00287	-1.93	0.0899	

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AD	-0.00914	-0.00457	0.00287	-1.59	0.15	
D	0.00683	0.00342	0.00287	1.19	0.268	
C	0.00666	0.00333	0.00287	1.16	0.279	
BD	-0.00566	-0.00283	0.00287	-0.988	0.352	
CD	-0.00473	-0.00236	0.00287	-0.824	0.434	
AC	0.00401	0.002	0.00287	0.699	0.504	
AB	3.38e-05	1.69e-05	0.00287	0.00589	0.995	

11.2 Response-surface coefficients — step yield

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.917	0.00651	141	4.36e-22	***
A	-0.0307	0.00444	-6.92	1.05e-05	***
B	-0.00757	0.00444	-1.71	0.112	
C	-0.00358	0.00444	-0.807	0.434	
D	-0.00548	0.00444	-1.24	0.239	
AB	0.00803	0.00471	1.71	0.112	
AC	0.000269	0.00471	0.0571	0.955	
AD	-0.000697	0.00471	-0.148	0.885	
BC	-0.00646	0.00471	-1.37	0.193	
BD	0.00185	0.00471	0.393	0.701	
CD	-0.00658	0.00471	-1.4	0.185	
A ²	-0.0103	0.0117	-0.882	0.394	
B ²	-0.00278	0.0117	-0.237	0.816	
C ²	0.000575	0.0117	0.049	0.962	
D ²	0.0108	0.0117	0.924	0.372	

Appendix D — Analytical methods summary

Table 11.5: Analytical methods, techniques and validation references (performance characteristics per the cited method validation reports).

Method	Technique	Analytes / attributes	Validation ref.	Reportable
Size variants	SEC-HPLC	Aggregate (HMW), monomer	AMV-3011	% HMW
Host-cell protein	ELISA	HCP	AMV-3012	ng/mg
Residual DNA	qPCR	Host-cell DNA	AMV-3014	ng/dose
Leached Protein A	ELISA	Leached Protein A	AMV-3016	ppm